



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



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Acronyms used in this material

- AI: Artificial Intelligence
- AUC: Area Under the (ROC) Curve
- BERT: Bidirectional Encoder Representations from Transformers
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAR: Cumulative Abnormal Return
- CI: Confidence Interval
- DELTA: Deep Learning for Text Analytics (Quantinar course)
- ECB: European Central Bank
- FNSPID: Financial News and Stock Price Integration Dataset
- GI: (Harvard) General Inquirer, dictionary IV-4
- IV-4: Harvard IV-4 (version 4 of the General Inquirer word lists)
- LDA: Latent Dirichlet Allocation
- LLM: Large Language Model
- LM: Loughran-McDonald (financial sentiment dictionary)
- OLS: Ordinary Least Squares
- ROC: Receiver Operating Characteristic (curve)
- SE: Standard Error
- TF-IDF: Term Frequency - Inverse Document Frequency
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Score a text with a sentiment dictionary and with the label probabilities of a large language model (LLM)
- Read a confusion matrix: accuracy, precision, recall, F1 and macro-F1
- Compare two classifiers with a paired test and detect training-data leakage and look-ahead bias
- Turn headline sentiment into a trading signal with correct timing, clustered standard errors and costs

- Background: Lecture 15; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- **Part A:** eight pen-and-paper exercises (A1–A8)
- **Part B:** eight computational exercises with inference (B1–B8)
- **Part C:** one open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- You may use an AI assistant, but declare it in [AI_USE.md](#) and log the errors you catch in [AI_ERRORS.md](#)
 - ▶ C2 trains exactly this: find the errors in an AI answer about sentiment backtests
- Always ask: was the text public before the return I try to predict, and did the model already know the outcome?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (tone with two dictionaries), A3 (metrics from a confusion matrix), A5 (label probabilities and temperature), A7 (break-even cost)
- Part B: B1 (FinBERT vs LM, McNemar), B3 (training-data leakage), B5 (clustered standard errors), B7 (does the LLM remember the market?)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (majority class; model: A3), A8 (longer holding; model: A7)
- Part B: B2 (LLM vs FinBERT; model: B1), B4 (how many labels; model: B3), B6 (event study; model: B5), B8 (prompt sensitivity; model: B1)
- Part C: C1, does headline sentiment pay after costs?; C2, critique an AI answer on an LLM sentiment backtest

Setup: Python Environment and Data

-  **Open in Colab:** [chapter15_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`; `transformers` only to recompute the scores
- Labelled texts: Financial PhraseBank (4 846 news sentences, 2 264 with full agreement) [Malo et al. \(2014\)](#); Twitter Financial News, validation set (2 388 headlines)
- Dated headlines: FNSPID (Financial News and Stock Price Integration Dataset) [Dong, Fan & Peng \(2024\)](#), 106 820 headlines on 16 large US stocks; daily closes of the stocks and of SPY (an S&P 500 index fund)
- The scores of every model are precomputed in the tables of Chapter 15; Quantlets: [MFM_ch15_sem_partA](#), [partB](#), [partC](#)

Formulas Used Today (1/2): Scores and Metrics

- Dictionary tone: $\tau = (P - N)/(P + N)$, P , N : counts of positive and negative words; $\tau = 0$ if $P + N = 0$
 - ▶ LM: the Loughran–McDonald financial word lists [Loughran & McDonald \(2011\)](#); GI: the Harvard General Inquirer IV-4 list, a generic psychology dictionary
- LLM label probabilities: $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ (z_k : logit of answer word k , T : temperature); net score $s = p_{\text{pos}} - p_{\text{neg}}$
 - ▶ FinBERT: a BERT (Bidirectional Encoder Representations from Transformers) model fine-tuned for financial sentiment [Araci \(2019\)](#); Qwen2.5: open-weight LLMs [Qwen Team \(2024\)](#)
- Class k : precision = $\frac{TP_k}{TP_k + FP_k}$, recall = $\frac{TP_k}{TP_k + FN_k}$, $F1_k = \frac{2 \text{ prec} \cdot \text{rec}}{\text{prec} + \text{rec}}$; macro-F1 = $\frac{1}{3} \sum_k F1_k$
 - ▶ TP_k : texts of class k labelled k ; FP_k : other texts labelled k ; FN_k : texts of class k labelled otherwise

Formulas Used Today (2/2): Inference and Signals

- McNemar test [McNemar \(1947\)](#): n_{01} texts where A is wrong and B right, n_{10} the reverse; under H_0 , $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
 - Bootstrap CI (confidence interval): resample texts 2 000 times, take the 2.5% and 97.5% quantiles
- AUC (area under the ROC curve): probability that a random “up” case gets a higher score than a random “down” case; 0.5 = no information
- Panel regression $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$: r^e excess return over SPY (%), z standardised daily score, $h = 0, 1, 2$ days after the news day d
 - Standard errors (SE): ordinary least squares (OLS) or clustered by day or by stock [Petersen \(2009\)](#); bp: basis point, 0.01%
- Strategy: Newey–West t [Newey & West \(1987\)](#); annualised Sharpe ratio $\sqrt{252} \bar{r}/s_r$; break-even cost = mean gross return on days with positions / 4 (four trades a day)

A1: One Sentence, Two Dictionaries [Solved]

Task

- Financial PhraseBank sentence, labelled **positive** by all annotators:
 - ▶ “Uponor improved its performance in spite of the decrease in residential construction in the US.”
- (a) Find the LM and the GI positive and negative words (15 tokens)
- (b) Compute the tone τ and the class for each dictionary
- (c) Why does the generic dictionary fail here?

Solution

- LM: positive *improved*; negative –
 - ▶ $\tau_{LM} = (1 - 0)/(1 + 0) = +1.00$: positive, correct
- GI: positive –; negative *spite*, *decrease*
 - ▶ $\tau_{GI} = (0 - 2)/(0 + 2) = -1.00$: negative, wrong
- GI misses *improved* and counts *spite* and *decrease*: they describe the housing market, not the firm
- A bag of words cannot see that “in spite of” reverses the role of what follows

A2: A Neutral Sentence [Proposed]

Solution discussed in class

Task

- Financial PhraseBank sentence, labelled **neutral** by all annotators:
 - ▶ “The ECB can mainly target inflation.”
- (a) LM and GI words, tone and class
- (b) Which dictionary agrees with the annotators, and why?
- (c) Is *inflation* bad news for a bank? For a bond holder?
- Model: A1

A3: Metrics from a Confusion Matrix [Solved]

Task

- FinBERT on the Twitter validation set:

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

- (a) Accuracy
- (b) Precision, recall and F1 for each class
- (c) Macro-F1; which class is the weakest, and why?

Solution

- Accuracy = $(264 + 1\,184 + 284) / 2\,388 = 72.5\%$
- Negative: prec $264 / 535 = 49.3\%$, rec $264 / 347 = 76.1\%$, F1 59.9%
- Neutral: prec 85.3% , rec 75.6% , F1 80.2%
- Positive: prec $284 / 465 = 61.1\%$, rec 59.8% , F1 60.4%
- Macro-F1 = $(59.9 + 80.2 + 60.4) / 3 = 66.8\%$, below the accuracy
- Negative is weakest in precision: 213 neutral headlines are called negative; a short strategy would trade on them

A4: The Loughran–McDonald Classifier [Proposed]

Solution discussed in class

Task

- The LM dictionary on the same headlines:

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

- (a) Accuracy, precision, recall and F1 per class, macro-F1
- (b) Which class does LM miss most, and why?
- Model: A3

A5: From Label Probabilities to a Score [Solved]

Task

- Qwen2.5-7B, prompt P1, one Twitter headline labelled **neutral** by the annotators
 - ▶ $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (rounded)
- (a) Predicted label and net score s ; is the label right?
- (b) Logit gap $z_{\text{neu}} - z_{\text{neg}}$
- (c) Probabilities, label and score at temperature $T = 2$

Solution

- Label $\arg \max_k p_k = \text{negative}$: wrong;
 $s = 0.000 - 0.608 = -0.608$
- $z_{\text{neu}} - z_{\text{neg}} = \ln(p_{\text{neu}}/p_{\text{neg}}) = \ln(0.392/0.608) = -0.437$
- At T : $p_k(T) \propto p_k^{1/T}$, here $\sqrt{p_k}$: 0.780, 0.626, 0; sum 1.406
 - ▶ $p(T) = (0.554, 0.446, 0.000)$, $s(T) = -0.554$
- Temperature is monotone: the label does not change, but the score shrinks towards 0; thresholds on s then select different headlines

A6: The Majority-Class Benchmark [Proposed]

Solution discussed in class

Task

- Twitter validation set: 1 566 neutral, 475 positive, 347 negative headlines ($N = 2\,388$)
- A “classifier” always answers *neutral*
- (a) Its accuracy
- (b) Precision, recall and F1 of each class; macro-F1
- (c) Which yardstick would you report for the LM classifier of A4, and why?
- Model: A3

A7: The Break-Even Cost of a Daily Signal [Solved]

Task

- FinBERT strategy of Lecture 15: at the close of $d + 1$, buy stocks with positive and sell those with negative news score of day d , each hedged with SPY; close one day later
- Days with positions: 96.2%; mean gross excess return on those days: 3.40 bp
- (a) Trades per day with positions; break-even cost per trade
- (b) Mean net return per day with positions at 5 bp per trade
- (c) How much larger would the gross edge have to be?

Solution

- Four trades: stock and SPY, in and out; break-even $c^* = 3.40/4 = 0.85$ bp
- Net: $3.40 - 4 \cdot 5 = -16.60$ bp per day with positions
- At 5 bp the gross edge must reach 20 bp a day: $20/3.40 = 5.9$ times more
- Daily turnover is the enemy: costs are paid every day, the edge is a few basis points

A8: Holding for Five Days [Proposed]

Solution discussed in class

Task

- ▣ Same edge as A7, 3.40 bp per day, but each position is held 5 days: one round trip (four trades) every 5 days
- ▣ (a) Cost per day at 5 bp per trade; break-even cost per trade
- ▣ (b) Is it realistic to assume the same daily edge on days $d + 2$ to $d + 6$?
- ▣ Model: A7

B1: Is FinBERT Better than LM? [Solved]

Task

- Twitter validation set, 2 388 headlines; labels of the LM dictionary and of FinBERT
- (a) Accuracy of each with a bootstrap 95% CI
- (b) Difference FinBERT – LM with a paired bootstrap CI (the same resampled texts for both)
- (c) Exact McNemar test [McNemar \(1947\)](#)
- Code: `b_pairs` in the notebook

Interpretation

- Why is the paired test more powerful than comparing the two intervals?
- Which texts carry no information in McNemar's test?
- Is a difference of this size useful for trading?

B1: Results [Solved]

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

- n_{01} (LM wrong, FinBERT right) = 539, $n_{10} = 252$; under H_0 we expect 395.5 each; exact $p < 0.001$
- The large-sample version $(n_{01} - n_{10})^2 / (n_{01} + n_{10}) = 104.1$ vs χ_1^2 gives the same verdict
- Texts on which both are right or both wrong cancel out; pairing removes the variation of text difficulty
- A clear win, outside FinBERT's training data; pp: percentage points

B2: Is a 7-Billion-Parameter LLM Better than FinBERT? [Proposed]

Solution discussed in class

Task

- ▣ Same data; FinBERT against Qwen2.5-7B, zero-shot, prompt P1 (no Twitter labels used by either)
- ▣ (a) Accuracies with CI, the paired difference with CI
- ▣ (b) Exact McNemar test
- ▣ Model: B1

Interpretation

- ▣ Do the two intervals overlap? Does that settle the question?
- ▣ Qwen2.5-7B has far more parameters and is much slower: what would you still gain from it?

B3: Training-Data Leakage [Solved]

Task

- FinBERT was fine-tuned on Financial PhraseBank [Araci \(2019\)](#)
- (a) Accuracy and macro-F1 of FinBERT, Qwen2.5-7B, LM and MiniLM + logistic regression on
 - ▶ PhraseBank, full agreement (2 264); all of PhraseBank (4 846); Twitter (2 388)
- (b) Drop from PhraseBank to Twitter for each model
- MiniLM [Wang et al. \(2020\)](#): sentence embeddings; its logistic regression is trained on the Twitter training set

Interpretation

- Twitter headlines are harder for everyone: how do you separate “harder data” from “memory of the training set”?
- Why does the MiniLM model move the other way?

B3: Results [Solved]

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	-8.0	69.9

- Drop: all of PhraseBank minus Twitter; FinBERT CI on PhraseBank [88.1; 89.8], on Twitter [70.7; 74.4]
- Difference in differences: FinBERT loses 9.2 pp more than Qwen2.5-7B, which never saw PhraseBank labels: a measure of the leakage
- MiniLM is trained on Twitter: better at home, worse on PhraseBank; every model is best on its own training domain

 MFM_ch15_sem_partB

B4: How Many Labels Do You Need? [Proposed]

Solution discussed in class

Task

- ▣ Logistic regression on MiniLM embeddings and on TF-IDF features (term frequency–inverse document frequency: word and word-pair counts, down-weighted if frequent in all texts)
- ▣ Trained on random subsets of the Twitter training set ($n = 100$ to $9\,543$), 10 repetitions each; evaluated on the validation set
- ▣ (a) Smallest n at which each beats FinBERT (72.5%)
- ▣ Model: B3

Interpretation

- ▣ Why do embeddings start higher but grow more slowly?
- ▣ How many texts would you label for a new market, e.g. Romanian stock news?

B5: Reaction or Prediction, and Which Standard Errors? [Solved]

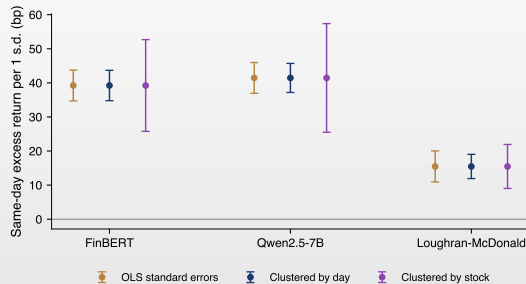
Task

- Panel of 18 282 stock-days with news (3 434 trading days, 16 stocks); daily score = mean of the headline scores
- (a) Regress r_{d+h}^e on z_d for FinBERT, Qwen2.5-7B and LM, $h = 0, 1, 2$
- (b) Standard errors: OLS, clustered by day, clustered by stock [Petersen \(2009\)](#)
- FNSPID gives only the date: day d is the first trading day on or after it, and $d + 2$ is the first return surely after publication

Interpretation

- Why do the conclusions for $h = 0$ and $h = 2$ differ?
- Which clustering matters here, and can you trust a clustered SE with 16 clusters?

B5: Results (1/2) [Solved]



- Slope b (bp of excess return per 1 standard deviation of score) at $h = 0$, with 95% intervals from the three standard errors

B5: Results (2/2) [Solved]

Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

- $h = 0$: large and significant under every SE: headlines describe the move of the day, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$): the tradable forecast is not there
- By day $\approx$ OLS (only 5.3 stocks per day); by stock the SE of FinBERT at $h = 0$ is 3.0 times larger: the slope differs across stocks
 - With 16 clusters the clustered SE is itself noisy: use t_{15} critical values or a wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#); Qwen at $h = 2$ ($t = +2.41$ by stock) is one of 9 tries

B6: Event Study with the Loughran–McDonald Tone [Proposed]

Solution discussed in class

Task

- Split news days into terciles of the daily LM tone; repeat with FinBERT
- For the most negative and the most positive third, average across stock-days of each date the excess returns on days -5 to $+5$
- (a) CAR (cumulative abnormal return) before the news (-5 to -1), day 0, day 1, after ($+2$ to $+5$), with Newey–West t
- Model: B5

Interpretation

- Does the LM tone separate winners from losers on day 0 as well as FinBERT does?
- Why is the post-event return positive for both groups? Tetlock (2007)

B7: Does the Model Remember the Market? [Solved]

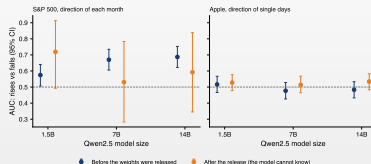
Task

- Prompt without context: “Did the S&P 500 index rise or fall during $\langle \text{month year} \rangle$?”; score $P(\text{rise})$ from the answer-word probabilities
- Qwen2.5 1.5B, 7B, 14B; weights published on 19 September 2024 [Qwen Team \(2024\)](#)
- (a) AUC for the direction of each month, before (to August 2024) and after (from October 2024) the release, with bootstrap CI
- (b) Permutation p -value for 14B before the release (2000 shuffles of the up/down labels)

Interpretation

- What does an AUC above 0.5 before the release mean, if the model sees no data?
- Can the post-release sample prove that the memory is gone?
- What does this imply for backtests that score old news with an LLM? [Glasserman & Lin \(2023\)](#)

B7: Results [Solved]



Qwen2.5	AUC before	CI 95%	AUC after	CI 95%	$\bar{P}(\text{rise})$ up / down
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]	0.36 / 0.30
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]	0.73 / 0.43
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]	0.81 / 0.54

- Before: 296 months (185 up, 111 down); after: 23 months; permutation p (14B) < 0.001
- 7B and 14B “know” the monthly direction of the past: no data were given, so the signal is memory of the training text
- After the release the intervals include 0.5, but 23 months are too few to prove the memory is gone

B8: Prompt Sensitivity [Proposed]

Solution discussed in class

Task

- Three prompts that ask the same question, Qwen2.5-7B and 14B, Twitter validation set
 - ▶ P1: “classify the sentiment . . . for investors”; P2: “good, bad or neutral for the stock price”; P3: “what is the sentiment of this text”
- (a) Accuracy with CI and share of *neutral* answers for each prompt
- (b) McNemar P1 vs P3; share of headlines with the same label under all three prompts
- Model: B1

Interpretation

- If you picked the best prompt on this set, what would you be doing?
- Why does the best prompt also answer *neutral* most often?

C1: Does Headline Sentiment Pay after Costs?

- Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, equal weights; close at $d + 2$
 - ▶ Days without news: no position, return 0; costs: 5 bp per trade, four trades a day
- (a) Mean daily gross return, Newey–West t , block-bootstrap 95% CI (blocks of 10 days)
- (b) Sharpe ratio before and after costs; break-even cost; Sharpe for costs of 0 to 10 bp
- (c) Strong news only: trade only if $|s_{i,d}| > 0.5$; (d) the same for Qwen2.5-7B and LM
- Discuss: timing, look-ahead of the LLM (B7), survivorship of the 16 stocks, crowding [Lopez-Lira & Tang \(2026\)](#)
- Deliverable: a two-page note with charts, and a Quantlet folder (code, Metainfo, charts)

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Answer of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of Lecture 15)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

□ Conclusion of the AI assistant

- ▶ "Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well."
- ▶ "Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day's headlines forecast the stock's excess return."
- ▶ "Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample."

□ The answer contains **three** errors typical for sentiment backtests; the code runs and the numbers are right

- ▶ Find each error: which line or sentence, and why it is wrong
- ▶ Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test
- ▶ Give the corrected code or statement
- ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Common Pitfalls

- ▣ Reporting accuracy without the majority-class benchmark and macro-F1
- ▣ Evaluating a model on its own training data (FinBERT on PhraseBank)
- ▣ Comparing two classifiers through separate intervals instead of a paired test
- ▣ Matching a date-only headline to the same-day return and calling it a forecast
- ▣ Backtesting an LLM on news older than the release of its weights
- ▣ Choosing the prompt, the threshold or the model after seeing the test results
- ▣ Ignoring costs and the number of clusters: four trades a day, only 16 stocks

Take-Home Assignment (Team Project)

- Choose a text source not covered in class: BVB (Bucharest Stock Exchange) announcements, Romanian financial news, central-bank statements or crypto posts
- Label at least 200 texts yourselves (two annotators, report the agreement)
- Compare LM, FinBERT, an open LLM and embeddings + logistic regression: accuracy and macro-F1 with CI, McNemar tests
- Build a signal with correct timing; report the period after the LLM's release separately
- Declare AI use in [AI_USE.md](#) and log every AI error you catch in [AI_ERRORS.md](#) (at least three per project, as in C2)
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

-  Natural Language Processing
-  The Next Word Problem
-  DELTA
-  LDA Extensions

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_15

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